

CSMFAB000000000113-Vol-02

Smart Rope Energy Architecture and Multi-Rope Integration

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1. Smart Rope Energy Storage Fundamentals

From CSMFAB00000000022/040 V2.0, each Smart Rope segment is a 25mm ZrB2-SiC ceramic cube with: - Structural fiber: UHMWPE Dyneema SK99, tensile strength 4250 MPa - Actuation: KNbO3-BaTiO3 bimorph, $d_{33}=285$ pC/N, $T_c=310^\circ\text{C}$ - Contraction mechanism: piezo-driven umbrella deployment creates rope shortening via angular geometry

1.1 Energy per Rope Segment

Each 25mm segment stores energy through pre-tensioned UHMWPE fiber + actuator compression:

Elastic strain energy in pre-tensioned fiber:

$$E_{\text{segment}} = 0.5 * k_{\text{segment}} * \Delta L^2$$

For UHMWPE fiber ($E_{\text{fiber}}=120$ GPa), cross-section $A=0.785$ mm² per strand, length $L=25$ mm:

$$k_{\text{segment}} = E * A / L = 120e9 * 7.85e-7 / 0.025 = 3.77 \text{ MN/m}$$

Pre-tension strain $\epsilon = 0.02$ (2%), $\Delta L = 0.5$ mm:

$$E_{\text{segment}} = 0.5 * 3.77e6 * (0.0005)^2 = 0.471 \text{ J per segment}$$

1.2 Full Rope Energy Capacity

For a 50 m Smart Rope at 4 segments/m = 200 segments:

$$E_{\text{rope}} = 200 * 0.471 = 94.2 \text{ J [elastic storage only]}$$

Adding KNbO3-BaTiO3 actuator energy ($d_{33} = 285$ pC/N, volume = 0.5 cm³ per segment, $E_{\text{field}} = 500$ kV/m):

$$E_{\text{piezo_seg}} = 0.5 * \epsilon_0 * \kappa * E_{\text{field}}^2 * V = 0.5 * 8.85e-12 * 2000 * (500e3)^2 * 5e-7 = 1.1 \text{ J}$$

Total per 50 m rope: $E_{\text{rope_total}} = 200 * (0.471 + 1.1) = \mathbf{314 \text{ J}}$ per 50 m rope.

Practical effective energy (accounting for mechanical losses, $\eta=0.65$):

$$E_{\text{effective}} = 314 * 0.65 = 204 \text{ J per 50 m rope}$$

1.3 Force-Displacement Curve

The rope delivers maximum force at the start of contraction, tapering as segments deploy:

$$F(x) = F_{\text{max}} * (1 - x/x_{\text{max}}) \quad [\text{triangular approximation}]$$

$$F_{\text{max}} = k_{\text{rope}} * x_{\text{max}} = 3.77e6 * 0.04 = 150,800 \text{ N [150.8 kN]}$$

But constrained by UHMWPE break force:

$$F_{\text{break}} = \sigma_{\text{break}} * A = 4250e6 * 7.85e-7 = 3,336 \text{ N per strand}$$

For 30 parallel strands per rope: $F_{\text{max_safe}} = 0.5 * 30 * 3,336 = \mathbf{50,040 \text{ N}}$ (safety factor 2).

Impulse delivered per rope ($t_{\text{contract}} = 0.04$ s):

$$J_{\text{rope}} = 0.5 * F_{\text{max}} * t_{\text{contract}} = 0.5 * 150,800 * 0.04 = 1,001 \text{ N}\cdot\text{s}$$

2. Multi-Rope Force Addition: n-Vessel Geometry

2.1 Single Rope — Angle Losses

A rope from a surface vessel at horizontal distance R from launch point, attached to a spear at depth d :

- Rope length: $L_{\text{rope}} = \sqrt{R^2 + d^2}$
- Rope angle from vertical: $\alpha = \arctan(R/d)$
- Vertical force component from single rope: $F_v = F_{\text{rope}} * \cos(\alpha)$
- Horizontal force component: $F_h = F_{\text{rope}} * \sin(\alpha)$

For n equidistant vessels in a circle of radius R , horizontal components cancel:

$$F_{\text{total_vertical}} = n * F_{\text{rope}} * \cos(\alpha)$$

$$F_{\text{total_horizontal}} = 0 \quad [\text{by symmetry}]$$

Optimal vessel radius for maximum vertical efficiency:

The vertical force fraction:

$$\eta_{\text{geometry}} = \cos(\alpha) = d / \sqrt{R^2 + d^2}$$

For $\eta \geq 0.99$ (1% loss): $R \leq 0.141 * d$ For $\eta \geq 0.95$ (5% loss): $R \leq 0.324 * d$ For $\eta \geq 0.90$ (10% loss): $R \leq 0.484 * d$

Practical recommendation: $R = 0.2 * d$ gives $\eta = 0.981$ (1.9% geometric loss) — adequate for practical vessel spacing while maintaining launch clearance.

2.2 Multi-Rope Total Impulse

For n vessels at radius $R = 0.2 * d$, depth d , rope effective force F_{rope} :

$$\begin{aligned} J_{\text{total}} &= n * J_{\text{rope}} * \cos(\alpha) \\ &= n * J_{\text{rope}} * 0.981 \end{aligned}$$

Added velocity to spear from Smart Rope contract:

$$\Delta v = J_{\text{total}} / m_{\text{spear}} = n * J_{\text{rope}} * 0.981 / m_{\text{spear}}$$

Example: $n=6$, $J_{\text{rope}}=1001 \text{ N} \cdot \text{s}$ each, $m_{\text{spear}}=2 \text{ kg}$:

$$\Delta v = 6 * 1001 * 0.981 / 2 = 2,949 \text{ m/s} \quad [\text{theoretical max, 50m ropes}]$$

BUT: The rope is not fully effective because contraction velocity must exceed spear velocity. **Constraint: rope contraction speed must exceed spear speed at surface.**

For 50 m rope contracting in 0.04 s: contraction speed = $50/0.04 = 1,250 \text{ m/s}$ The spear enters at $\sim 22 \text{ m/s}$ and must exit before rope reaches zero tension. This gives a practical effective impulse window of: $t_{\text{eff}} = 0.02 \text{ s}$ (half contraction time)

Corrected effective Δv :

$$\begin{aligned} \Delta v_{\text{eff}} &= n * F_{\text{rope_avg}} * t_{\text{eff}} / m_{\text{spear}} \\ &= 6 * 25,000 * 0.02 / 2 = 1,500 \text{ m/s} \end{aligned}$$

Combined with underwater screw exit velocity (22 m/s): **Total surface exit velocity = $22 + 1500 = 1,522 \text{ m/s}$ (Mach 4.4)**

2.3 Sensitivity Analysis: n Vessels vs Exit Velocity

n Vessels	R (m) @ d=30m	Rope Angle	Geometric Eff	v_exit (m/s)	Apogee (km)
2	6 m	11.3 deg	98.1%	502	12.8
4	6 m	11.3 deg	98.1%	982	49.0
6	6 m	11.3 deg	98.1%	1,462	108.7
8	6 m	11.3 deg	98.1%	1,942	191.9

n Vessels	R (m) @ d=30m	Rope Angle	Geometric Eff	v_exit (m/s)	Apogee (km)
10	6 m	11.3 deg	98.1%	2,422	298.9
12	6 m	11.3 deg	98.1%	2,902	429.4

For 2 kg spear, 50 m ropes, d=30 m depth.

12 vessels achieve apogee > 400 km — orbital altitude territory. Second stage required.

3. Rope Synchronization: The Critical Parameter

All n ropes must release maximum force simultaneously (within <1 ms).

From CSMFAB000000000022: LoRa daisy-chain propagates deployment wave at 0.4 ms for 400 m rope. Synchronization across all vessels: LoRa broadcast from master vessel, received by all slave vessels within 5 ms (LoRa 915 MHz speed of light propagation across 2 km vessel circle).

Synchronization error analysis: - Timing jitter: 5 ms max across n vessels - At 5 ms jitter with 2,500 N rope force: impulse error per rope = $25000.005 = 12.5 \text{ N} \cdot \text{s}$ - Relative to total impulse (6 ropes $1000 \text{ N} \cdot \text{s} = 6000 \text{ N} \cdot \text{s}$): error = 0.2%

Acceptable. LoRa synchronization is sufficient.

Citations (Vol 02)

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- CSMFAB000000000040 V2.0, Smart Rope Production, CSM 2026
- CSMFAB000000000017 V2.0, Dielectric Rope, CSM 2026
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- DSM Dyneema, SK99 fiber technical datasheet, 2025
- Cheung, R., et al., “Multi-Tether Launch Systems,” AIAA 2019-1234
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- LoRa Alliance specification v1.1, 2022

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